

IB MATH AA SL

Lesson 34

Limits & the Derivative

Calculus · 60-minute lesson

Syllabus SL 5.1 · Property of Lynda Badmus Education

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Do Now — recall

ON YOUR WHITEBOARD

1. Find the gradient between (1, 2) and (3, 8).
2. What does 'gradient' measure?
3. Gradient of a horizontal line?

Answers (click to reveal)

1. $(8-2)/(3-1) = 3$
2. steepness / rate of change
3. 0

What we're learning today

LEARNING OBJECTIVES

- 1 Understand**
the derivative as a gradient function.
- 2 Interpret**
 $f'(x)$ as a rate of change.
- 3 Use**
derivative notation dy/dx and $f'(x)$.
- 4 Read**
the gradient at a point from $f'(x)$.

SUCCESS CRITERIA — I CAN...

- ✓ explain the derivative as the gradient of a curve
- ✓ use the notations dy/dx and $f'(x)$
- ✓ interpret $f'(x)$ as a rate of change
- ✓ find the gradient at a point once $f'(x)$ is known

The derivative

GRADIENT OF A CURVE

$$f'(x) = \frac{dy}{dx}$$

Gradient

the steepness of the curve at a point

Rate of change

how fast y changes as x changes

Notation

$f'(x)$ and dy/dx mean the same thing

FROM CHORD TO TANGENT

Average gradient = chord between two points.

Limit

bring the points together $\rightarrow$ the tangent's gradient.

That limit

is the derivative.

Interpret the derivative

QUESTION

For a curve $y = f(x)$, it is found that $f'(2) = 5$.

What does this tell us?

SOLUTION

$f'(2)$ is the gradient at the point where $x = 2$:

The curve is sloping upward there, gradient 5.

So y increases 5 units per 1 unit of x near $x = 2$.

Your turn — meaning

ANSWER IN YOUR BOOK

1. If $f'(3) = 0$, what is happening at $x = 3$?
2. If $f'(x)$ is negative, is the curve rising or falling?
3. Another name for dy/dx ?

Answers (click to reveal)

1. flat (stationary point)
2. falling
3. $f'(x)$ / the gradient function

Notation & rates

WAYS TO WRITE IT

$$\frac{d y}{d x} = f'(x)$$

dy/dx

'the rate of change of y with respect to x'

Sign

positive → increasing; negative → decreasing

IN CONTEXT

ds/dt

velocity (rate of change of distance)

dV/dt

rate a volume changes

Units

matter — e.g. m per s.

Gradient at a point

QUESTION

A curve has gradient function $f'(x) = 2x$.

Find the gradient at $x = 4$.

SOLUTION

Substitute $x = 4$ into $f'(x)$:

$$f'(4) = 2(4)$$

Evaluate:

$$f'(4) = 8$$

Exam-style question

A curve $y = f(x)$ has gradient function $f'(x) = 3x - 1$.

- (a) Find the gradient at $x = 2$. [2]
- (b) State what your answer means. [1]

MARK SCHEME — reveal after students attempt (tutor only)

- (a) $f'(2) = 3(2) - 1 = 5$ (M1)(A1)
- (b) the curve's gradient (rate of change) at $x = 2$ is 5 (A1)

Common mistakes



Derivative = gradient

$f'(x)$ gives the slope, not the y-value.



$f'(a)$ is a number

Substitute the x-value into $f'(x)$.



Sign meaning

$f'(x) > 0$ rising; < 0 falling; $= 0$ stationary.



Notation

dy/dx and $f'(x)$ are the same thing.

Mixed practice

QUESTIONS

1. $f'(x)=4x$. Find $f'(3)$.
2. If $f'(5)=0$, what's happening at $x=5$?
3. $f'(x)=2x-6$. Find $f'(1)$.
4. Is the curve rising if $f'(x)=-2$?

Answers (click to reveal)

1. 12
2. stationary (flat)
3. -4
4. no, falling

Plenary & exit ticket

KEY FORMULAS

$$f'(x) = \frac{dy}{dx}$$

$$f'(a) = \text{gradient at } x=a$$

EXIT TICKET — before you leave

1. $f'(x)=6x$. Find $f'(2)$.
2. $f'(x)<0$ means rising or falling?
3. Another symbol for $f'(x)$?

Answers (click to reveal)

1. 12
2. falling
3. dy/dx

Homework

SET ON DR FROST MATHS (auto-marked)

Task:

Limits & the derivative.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 35

Differentiating polynomials with the power rule.

$$\frac{d}{dx} x^n = n x^{n-1}$$